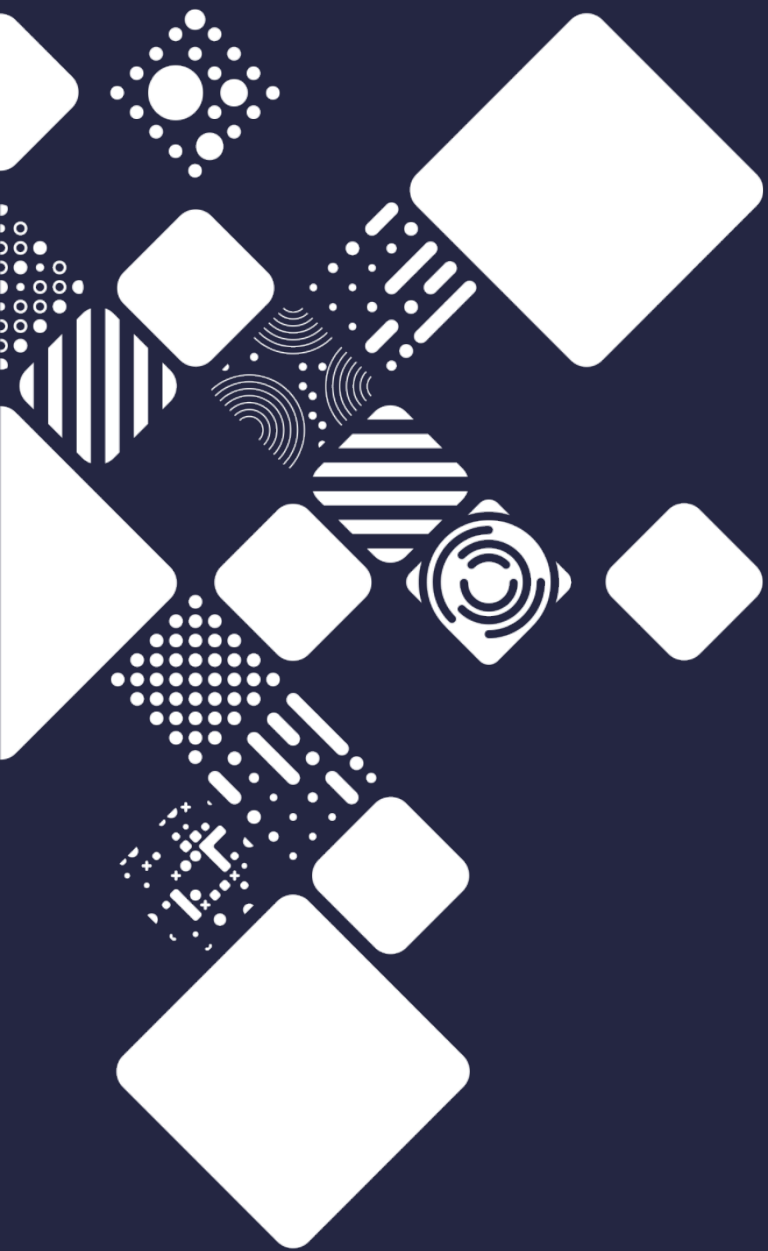




Decoding the spectrum of *RAD51C* variant effects via Saturation Genome Editing

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Adams Lab

RAD51C



Cancer Research UK



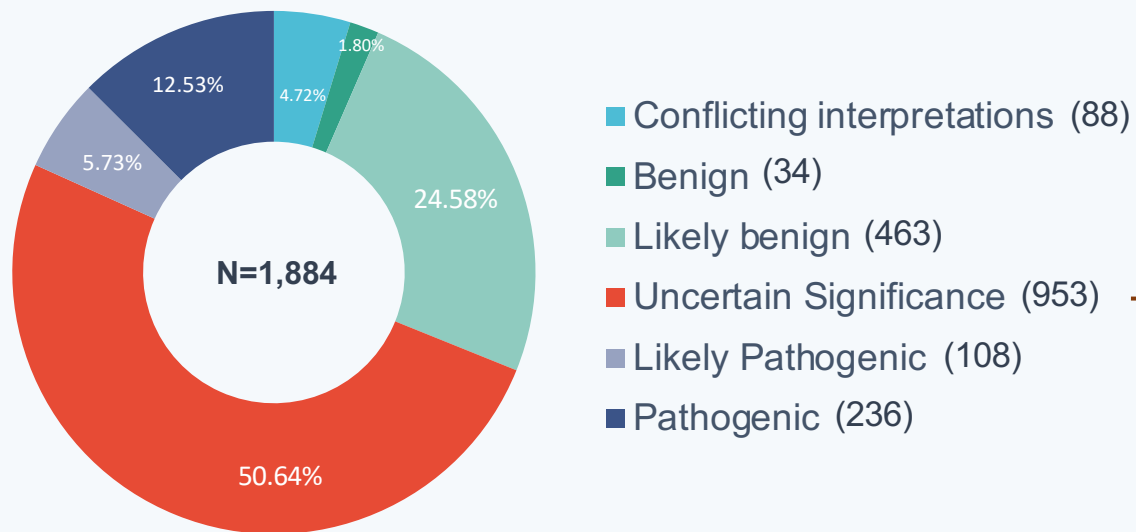
- Involved in homologous recombination
- Predisposition to breast and ovarian cancer
- Association with other cancers:
 - Gastric
 - Pancreatic
- Biallelic mutations can cause Fanconi Anemia
 - Susceptibility to other cancers (Leukaemia)



García-de-Teresa et al. (2020) Genes.

RAD51C variants reported in ClinVar

50% are classified as Variants of Uncertain Significance (VUS)



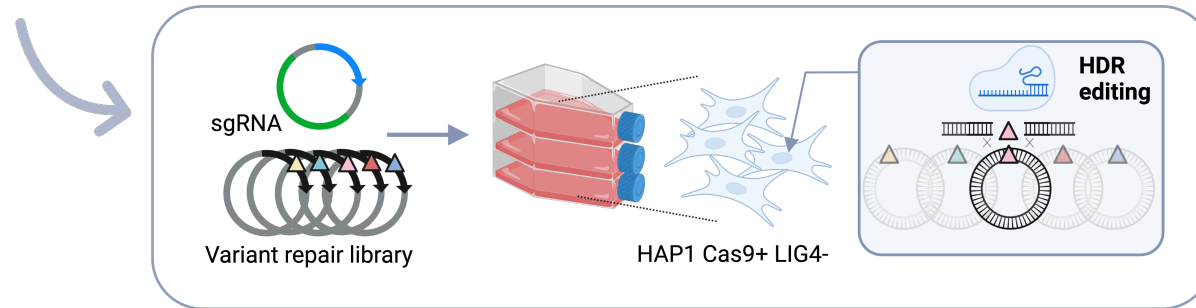
- Limits clinical interpretation.
- Complicates medical management of patients.

822 are missense variants

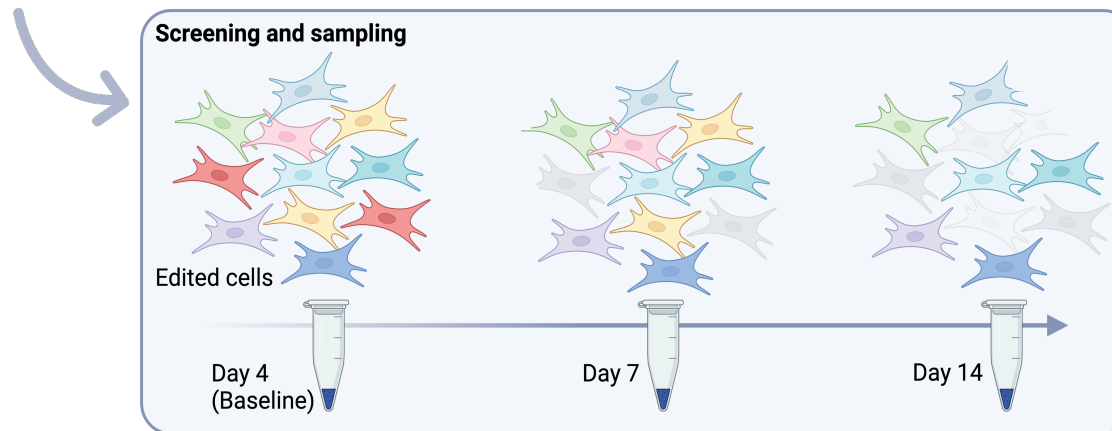
Saturation Genome Editing (SGE) of *RAD51C*



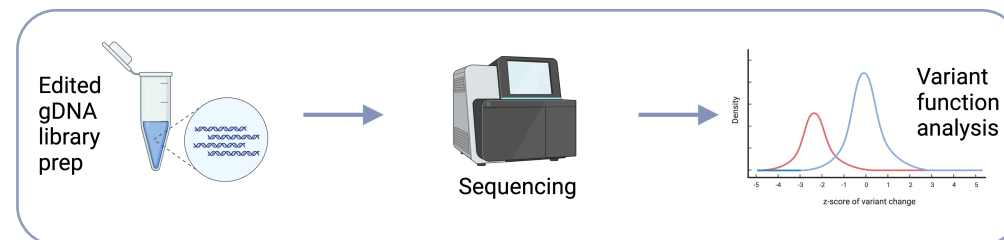
- SGE across 9 exons



- Three technical replicates

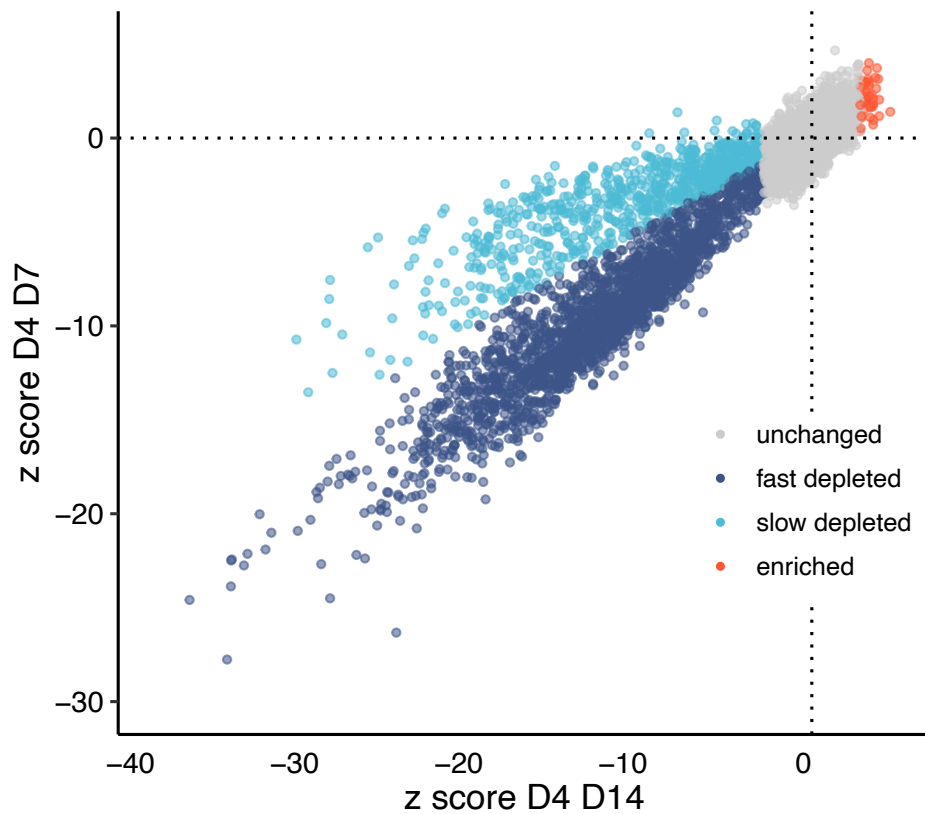


- Three timepoints



- 108 libraries sequenced

Functional evaluation of 9,188 *RAD51C* variants: Variant kinetics reveals distinct classifications of effect



Functional classifications:

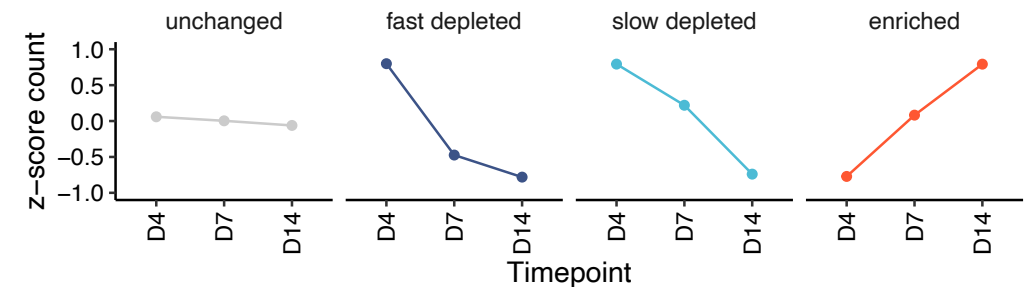
Unchanged $\rightarrow$ $\text{FDR} \geq 0.01$

Depleted $\rightarrow$ $\text{FDR} < 0.01$ & negative z-score D4D14

Enriched $\rightarrow$ $\text{FDR} < 0.01$ & positive z-score D4D14

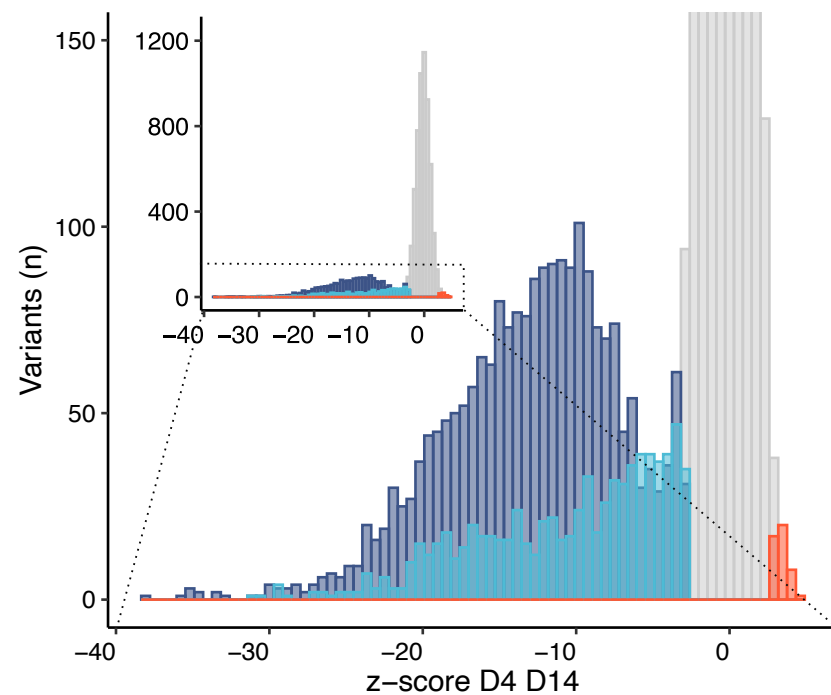
$\rightarrow$ Fast-depleted variants depleted by D7

$\rightarrow$ Slow-depleted not depleted until D14

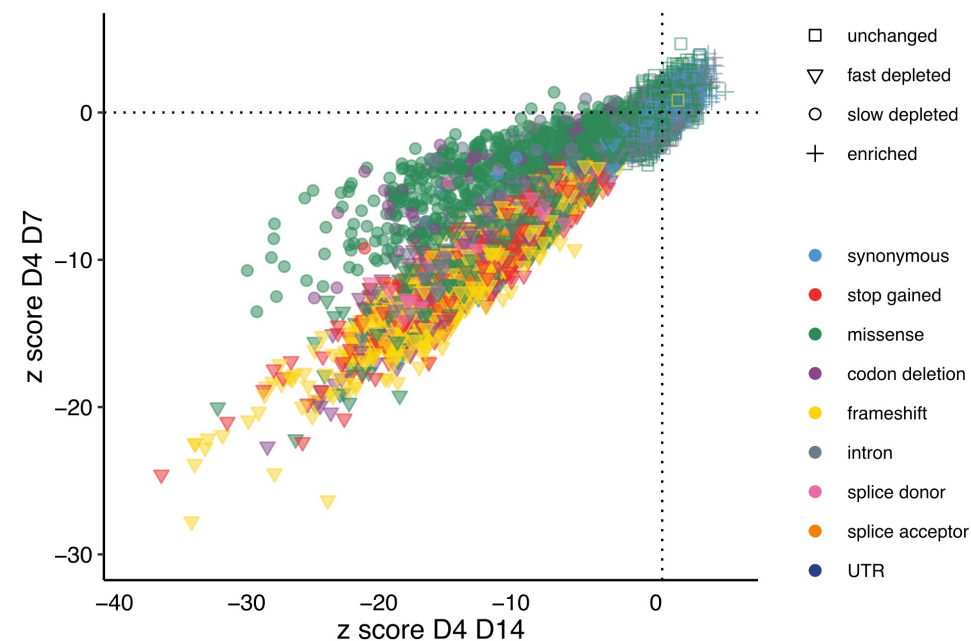




Functional classification of variant effect



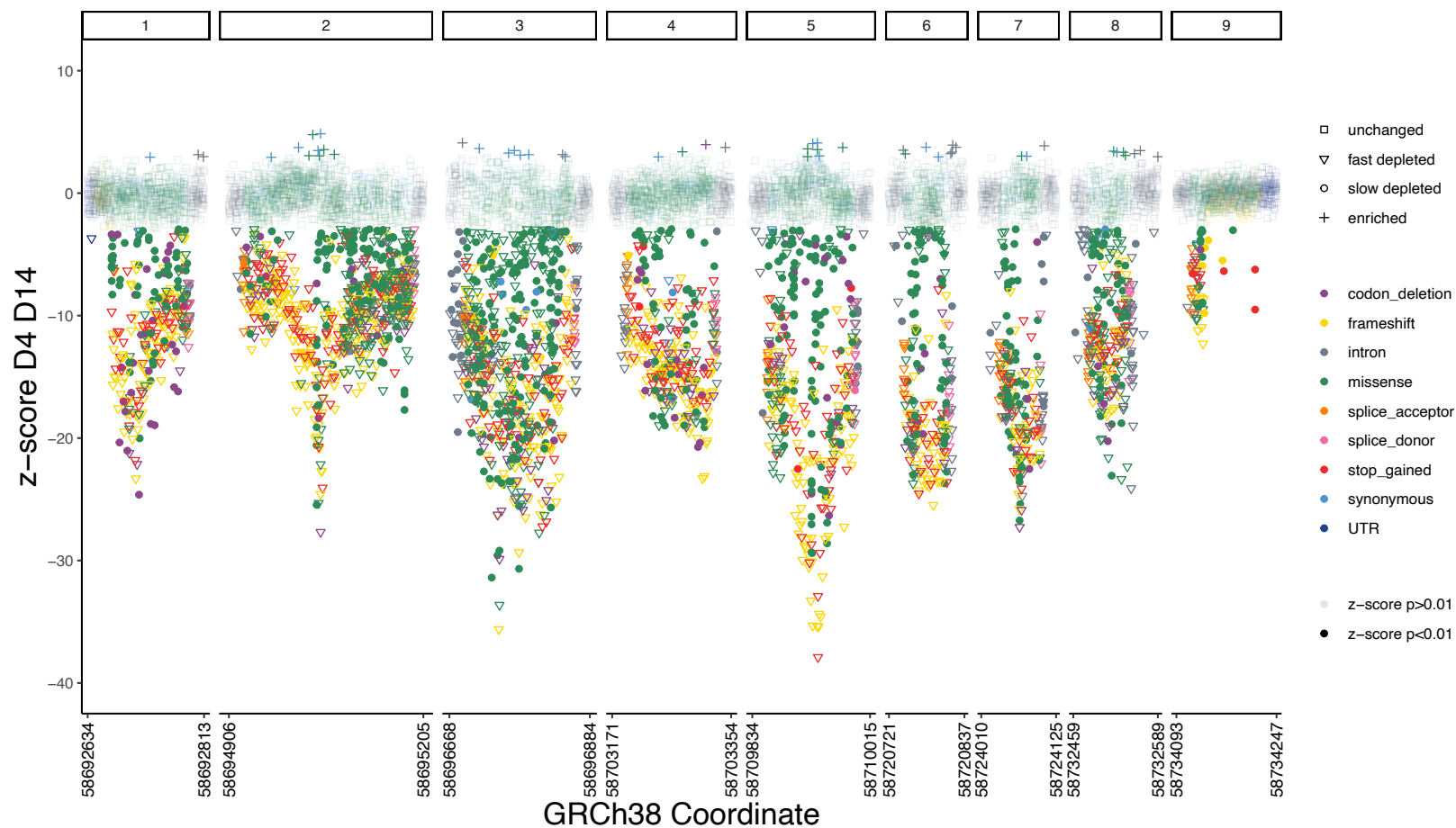
unchanged	➤	6,045
fast depleted	➤	2,254
slow depleted	➤	840
enriched	➤	49



Fast-depleted → predominantly stop gained and frameshift

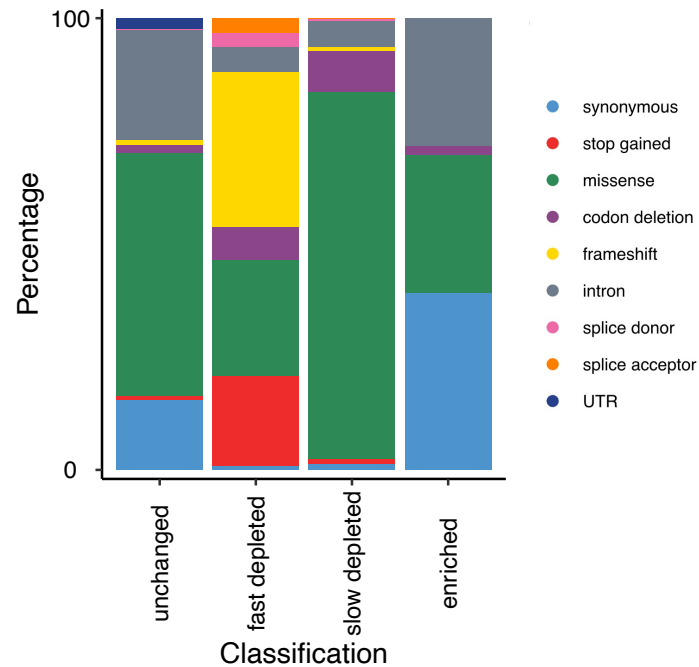
Slow-depleted → predominantly missense

Full nucleotide-level map of functional classifications



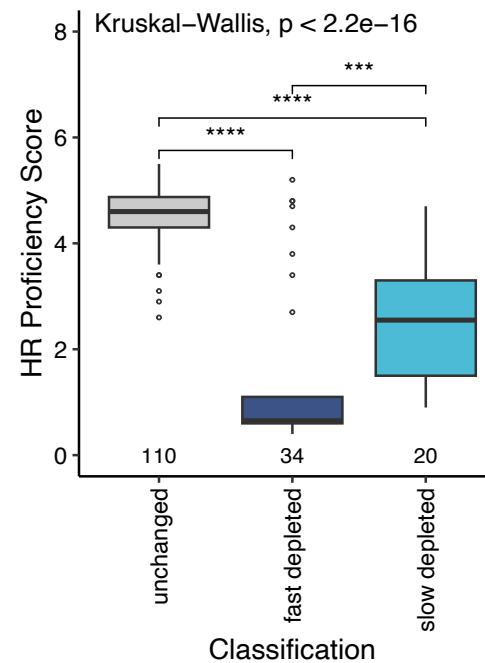
***RAD51C* missense variants**

Slow depleted variants:
predominantly missense



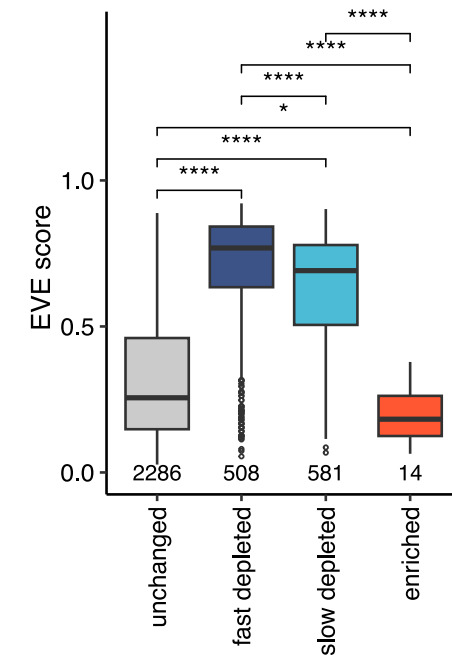
Slow depleted missense variants are
associated with moderate HR activity

Hu et al. (2023) *Cancer Res.*



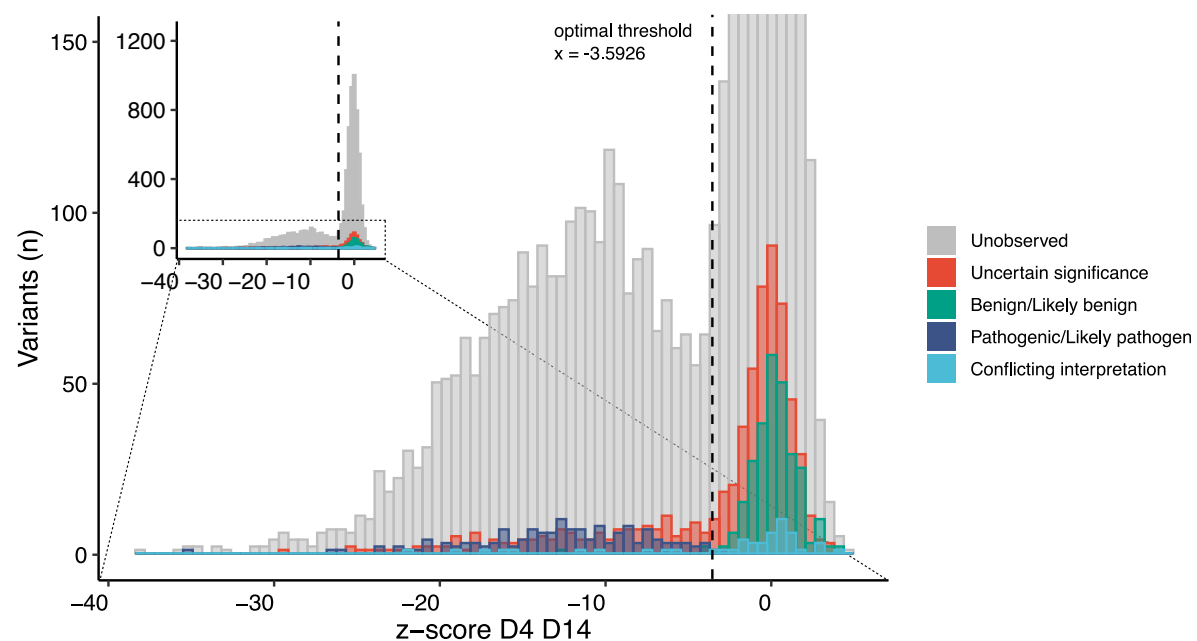
**Cell-viability phenotype assayed in SGE is
likely a direct readout of HR proficiency**

Slow depleted are less
conserved through evolution



SGE data are highly sensitive and specific for variant classification

Clinically observed variants in ClinVar



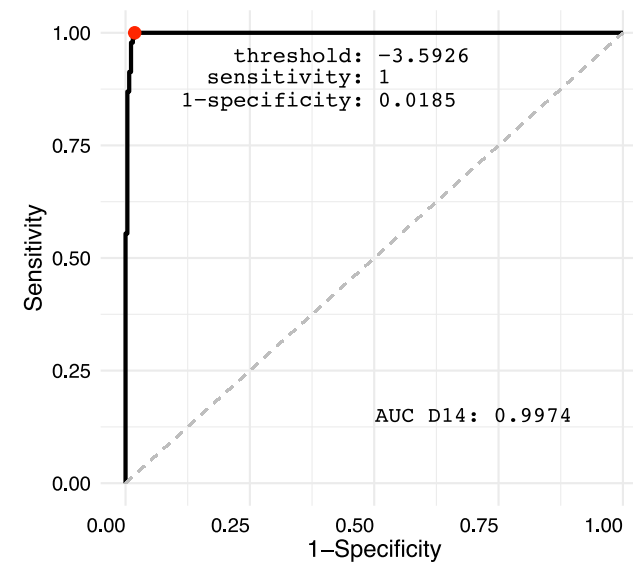
Known pathogenic variants deplete

Predictive power of SGE data

True positives (pathogenic) $n=92$

True negatives (benign) $n=271$

100% sensitivity, >98% specificity

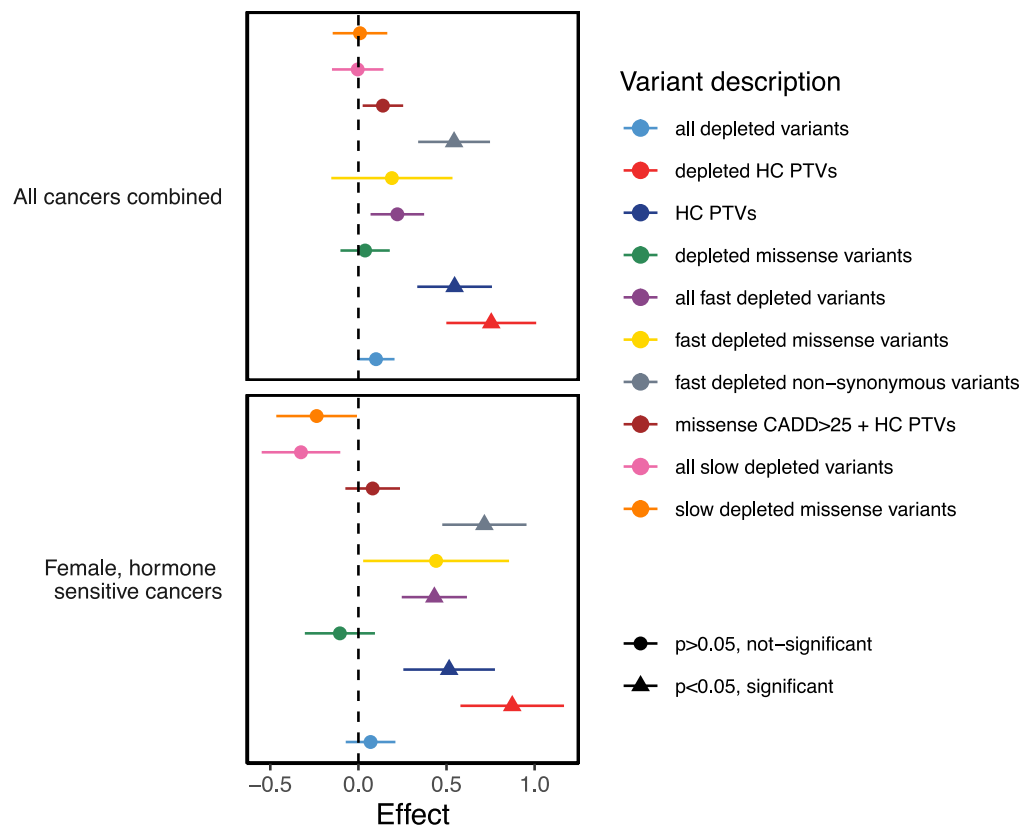


SGE depleted variants associated with cancer in UKBB data

WES data from 421,064 individuals: association between *RAD51C* and cancer only in females



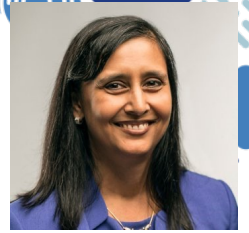
Yajie Zhao, MRC Epidemiology Unit



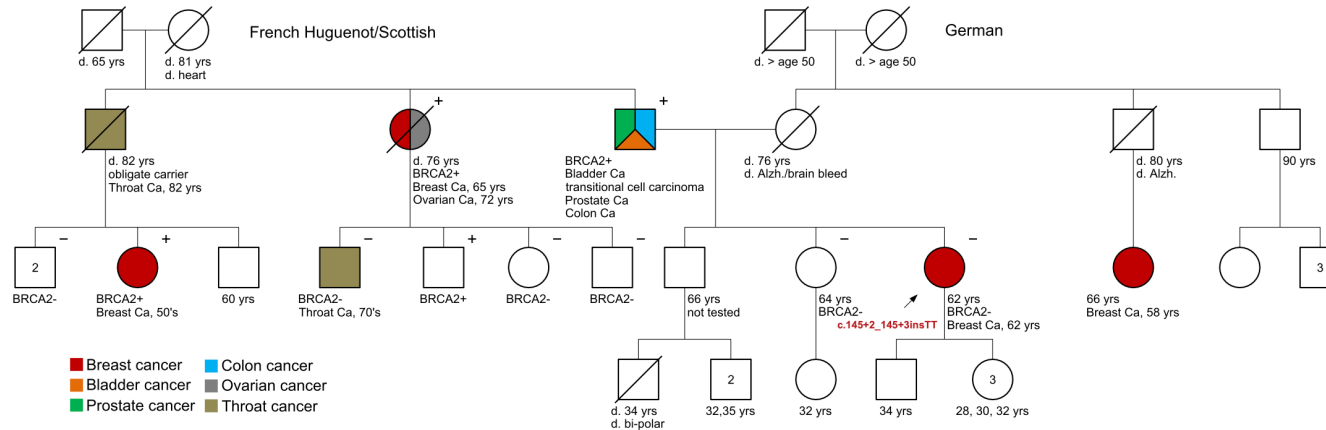
478 SGE variants identified

- 55 SGE-fast depleted (440 carriers)
- 39 SGE-slow depleted (39)
- 5 SGE-enriched (233)
- 379 SGE-unchanged (30,248)

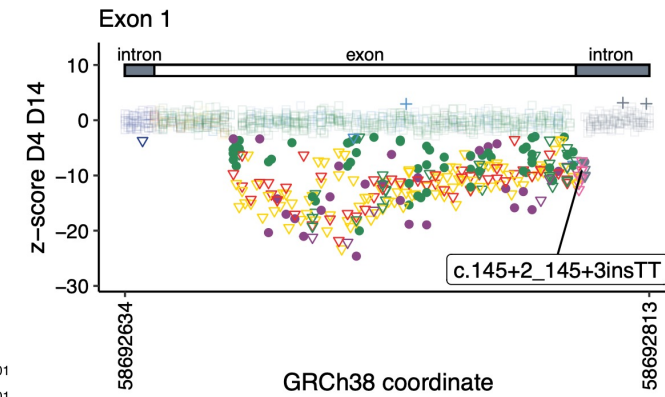
RAD51C VUS in cancer-segregating kindred



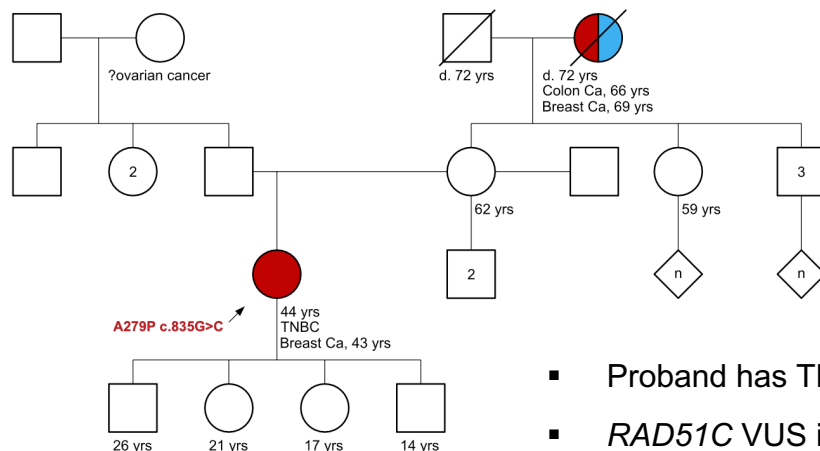
Tuya Pal, VUMC & ICARE



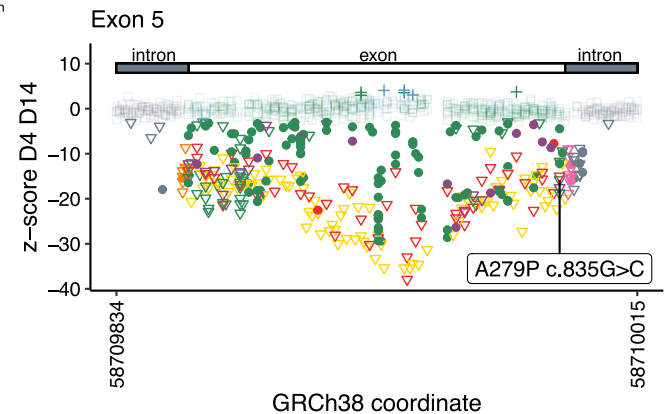
- Family with individuals *BRCA2*+
- Proband has BC and is *BRCA2*-
- RAD51C* intronic VUS identified: c.145+2_145+3insTT



➤ Fast-depleted

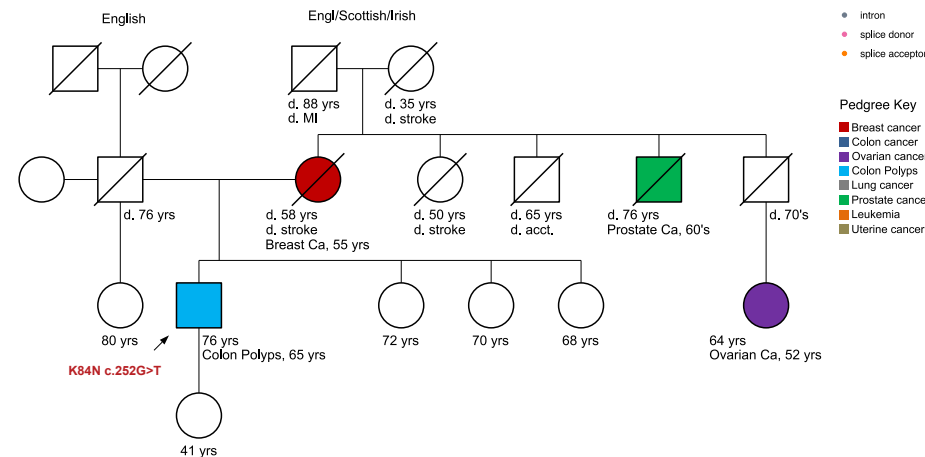
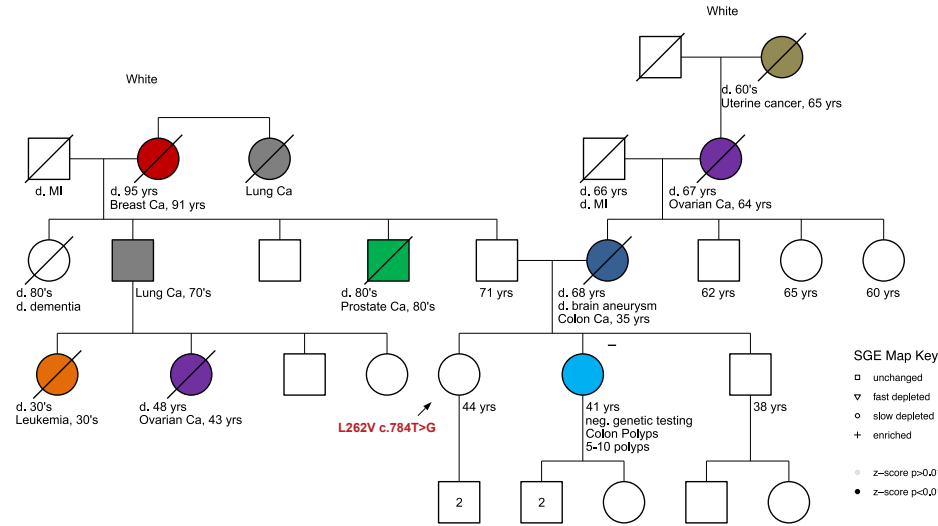
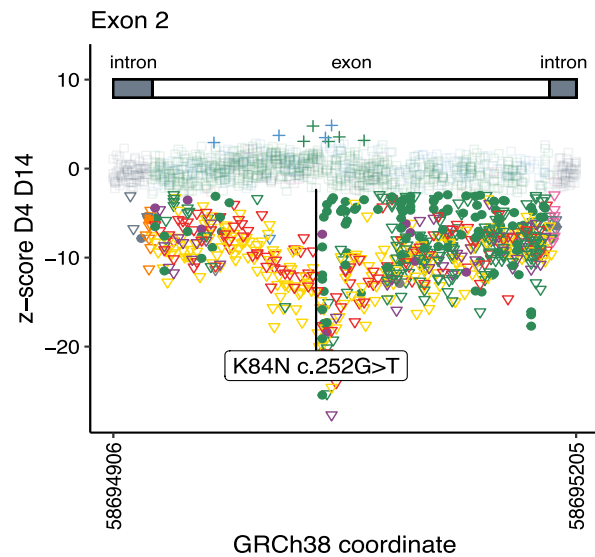
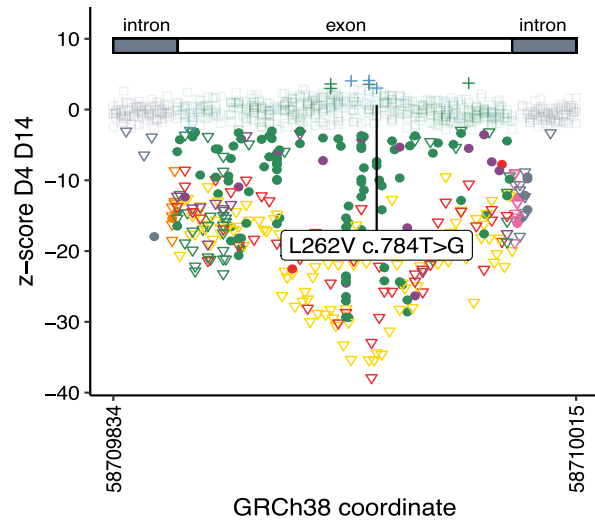


- Proband has TNBC
- RAD51C* VUS identified: p.A279P



➤ Fast-depleted

RAD51C VUS in cancer-segregating kindred



- Family history of cancer
- Proband without cancer diagnosis
- RAD51C VUS identified: p.L262V

➤ Both are unchanged

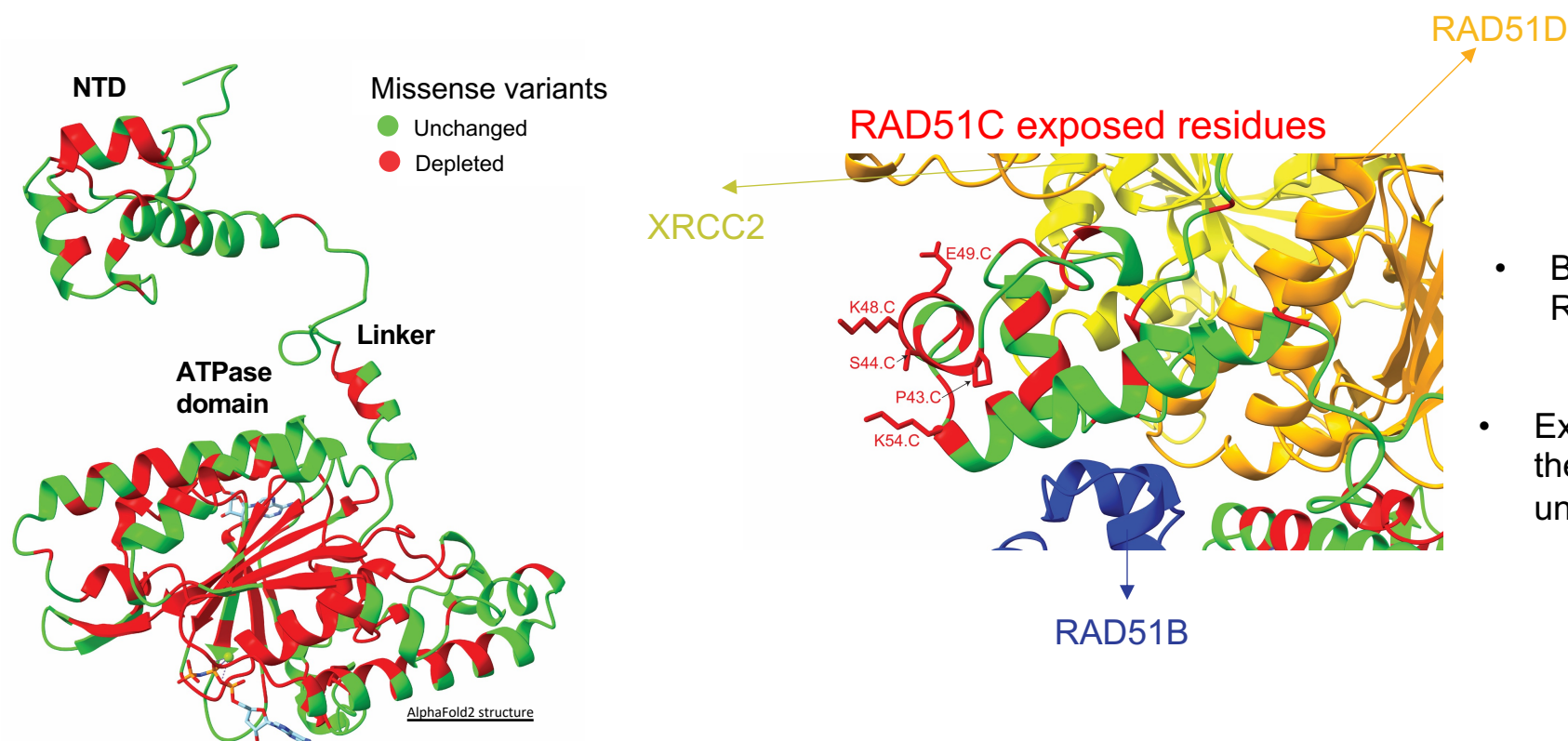
- Family history of cancer
- Proband has colon polyps
- RAD51C VUS identified: p.K84N



Maria Jasin and Fang Zhang, MSKCC

SGE identifies residues critical for RAD51C function

- Structural position suggests variant impact
- Structure alone mispredicts RAD51C variant effects.



- BCDX2 complex: (RAD51B, RAD51C, RAD51D and XRCC2).
- Exposed residues in the context of the BCDX2 complex: suggesting unidentified interactors.

- SGE depleted *RAD51C* missense variants map to approximately half of the *RAD51C* residues (175/376)

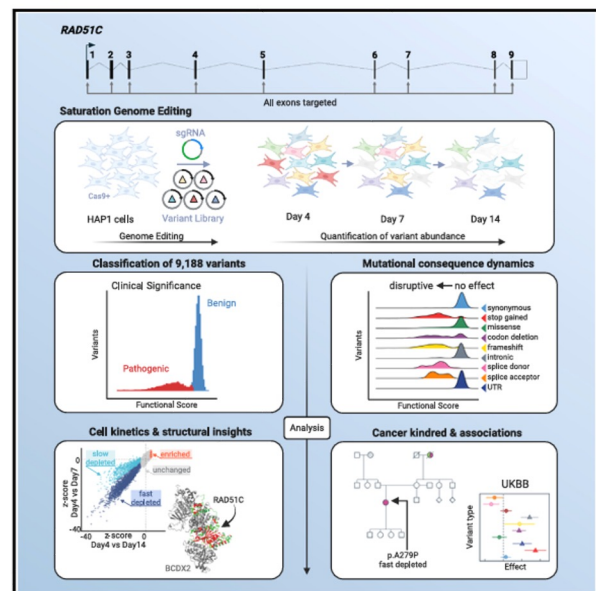
Summary

- We applied SGE to functionally classify 9,188 *RAD51C* variants.
- Slow-depleted variants may represent a group of pathogenic variants of reduced penetrance.
- Associations between SGE depleted variants and hormone cancers diagnoses.
- New residues required for the function of RAD51C.
- We provide a resource for *RAD51C* variant interpretation, aiding VUS reclassification and improving clinical management in cancer-prone families.

Cell

High-resolution functional mapping of *RAD51C* by saturation genome editing

Graphical abstract



Article

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In brief

Functionally compromising variants in *RAD51C* pre-dispose individuals to breast and ovarian cancer; identifying these variants is critically important for the clinical management of patients and families. A saturation genome editing approach functionally classifies >99% of *RAD51C* coding sequence single-nucleotide variants and defines a spectrum of variant effects across multiple mutation types and exon/intron sequences.



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